GGG-C-111C
February 12, 1987
SUPERSEDING
GGG-C-111B
October 14, 1966

FEDERAL SPECIFICATION

CALIPERS AND GAGES, VERNIER

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers vernier-type calipers and gages.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity as indicated by the title of the specification or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types. Calipers and gages shall be of the following types, classes, and styles, as specified (see 6.2):

Type I. Calipers, vernier.
  Class 1. English measure.
  Class 2. Metric measure.
  Class 3. English and metric measure.
Type II. Calipers, gear tooth, vernier.
  Class 1. English measure.
    Style A. 20 to 2 diametral pitch.
    Style B. 10 to 1 diametral pitch.
Type III. Gages, depth, vernier.
  Class 1. English measure.
  Class 2. Metric measure.
  Class 3. English and metric measure.
Type IV. Gages, height, vernier.
  Class 1. English measure.
  Class 2. Metric measure.
  Class 3. English and metric measure.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on the date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-P-40 - Packaging and Packing of Hand Tools

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Commercial Item Descriptions as outlined under General Information in the Index of Federal Specifications and Standards. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the General Services Administration, Specifications Unit (WFSIS), 7th and D Street SW, Washington, DC 20407.)

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.)

FSC 5210

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of calipers and gages otherwise conforming to this specification.

3.2 Materials. The beams or column slides and all parts having gaging contact with the work shall be of tool steel or tool steel with carbide inserts. Clamps, screws, and other parts where accuracy of the caliper or gage is not affected by wear shall be the manufacturer's commercial practice. Bases of type IV gages shall be of tool steel or ductile iron.

3.3 Design. The calipers and gages shall be of the manufacturer's latest design. All parts of the calipers and gages shall be new. All parts subject to wear, breakage, or distortion shall be easily replaceable.

3.3.1 Main-scale member. The main-scale member shall be graduated as hereinafter specified for each type, class, and style of instrument. The main-scale member shall be of such cross-sectional area as to insure stability without undue weight when the instrument is in use to measure dimensions at its maximum range. The measuring face and slide locating edge shall be straight within 0.0003 inch for each foot of member length, total tolerance shall not be accumulated within any foot increment, and any deviation in straightness shall be evenly distributed over the entire length of the instrument.

3.3.2 Slide. The slide shall be machined to receive the main-scale or complementary member and shall have such a fit and surface finish as to slide along the entire length of the member without lateral shake or bind when not locked in position. The locating side of the slot shall be square with the measuring contact plane within 0.0003 inch in the full length of the contact surface. Any deviation in squareness between the main-scale member and the locating side of the slide slot shall be in the toe-in direction. The side opposite the movable jaw shall be fitted with a nib and a locking screw.

3.3.3 Vernier plate. The vernier plate shall either overlap or be flush with the edge of the main scale. If the vernier plate overlaps the main scale, the overlap shall be not less than 1/64 inch with the edge of the plate straight within 0.002 inch in full length, and beveled to prevent parallactic error. If installed flush with the main scale, the edge of the vernier plate shall slide easily along the main scale, and graduations of both the main and vernier scales shall extend fully to the edges. The vernier plate shall be held in place by two or more screws, and the mounting holes shall be elongated to allow not less than 0.010 inch longitudinal adjustment.

3.3.4 Measuring contacts. The measuring contacts shall be as defined for each type and class of instrument. The contact surface shall have a surface finish of 8 microinches A.A. or better and a Rockwell hardness of C-60 to C-65.

3.3.5 Locking screws. The locking screws shall be of the fine-thread series, the head of which shall be medium diamond or straight knurled.

3.3.6 Fine adjustment clamp. Unless otherwise specified herein, a fine adjustment clamp shall be provided which shall clamp to the main-scale member while the slide remains movable.

3.3.7 Fine adjustment screw. Unless otherwise specified herein, a fine adjustment screw shall move the slide for fine positioning while the fine adjustment clamp is in a clamped condition. The fine adjustment screwnut or knob shall be knurled. All threads in the fine adjustment mechanism shall be of a fine-thread series.

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3.3.8 Scales and graduations. Calipers and gages shall have a main scale and a vernier scale. The two scales shall be graduated as hereinafter specified for each type, class, and style.

All vernier scales shall have one more graduation than its complementary main scale over a length on the main scale equivalent to the graduated length of the vernier scale. All graduations shall be machine cut or etched 0.003 to 0.006 inch in width and from 0.001 to 0.003 inch in depth and shall not vary in width on both the main and vernier scales of any one instrument by more than 0.001 inch. The edges of all graduations shall be straight within 0.0002 inch in the full length of the graduation and shall be sharp and well defined so as to facilitate use of the instrument under normal service conditions. In addition, all graduations shall be made to contrast with surrounding areas of the calipers by chemically etching the graduation lines or by other similar means so that graduations are distinct and easily readable. The main-scale graduations shall be parallel to each other and square with the locating edge of the main-scale member within 0.0004 inch in the full length of the graduation. The vernier-scale graduations shall be parallel to each other and square with the reading edge of the vernier plate within 0.0002 inch. Indexes of both the main and vernier scales shall be so located with reference to the measuring contact planes as to be direct reading.

3.3.9. English graduated instruments. All English graduated instruments except type II, class 1, style A caliper (gear tooth vernier 20 to 2 diametral pitch), shall be graduated to read in thousandths of an inch increments and shall be graduated as specified herein.

3.3.9.1 Main scale. All calipers and gages except type II, class 1, style A caliper shall be graduated as follows: Each inch of the main scale shall be represented by a graduation marked with the applicable inch increment figure. Each inch shall be subdivided into 10 increments of 0.100 inch. Each 0.100 inch increment shall be marked with the appropriate increment figure. Length of graduations and height of figures shall be such as to be easily readable.

Each 0.100 inch increment shall be further divided (depending on the type of vernier used), either into 4 increments of 0.025 inch (25 division instruments) or into 2 increments of 0.050 inch (50 divisions instruments). Each inch graduation within the first foot of main-scale member or less shall be accurately located with reference to the zero index within 0.0003 inch and within each additional foot of length an additional 0.0003 inch, total tolerance shall not be accumulated within any 1-foot increment for instruments of over one foot in length. All graduations within any 1-inch increment shall be accurately located within that inch increment within 0.0002 inch.

3.3.9.2 Vernier scales. Vernier scales for 25 division English graduated instruments shall have a graduated length of 0.6000 or 1.2250 inch ± not more than 0.0002 inch and shall be divided into 25 equal increments accurately spaced to within ± 0.0002. All graduations shall be of such lengths as to be easily readable with those graduations representing 0 and each 0.005 increment having extra length to render them easily distinguishable. The zero index and each graduation representing 0.005 increments shall be marked with the increment figure. Vernier scales that are divided into 50 increments shall have a graduated length of 1.225 or 2.450 ± not more than 0.0002 inch and shall be accurately spaced to within 0.0002 inch. Each instrument shall be marked with the reading increment.

3.3.10. Metric graduated instruments. All metric graduate instruments shall be graduated to read in "1/50-mm" increments and shall be graduated as specified in 3.3.10.1 and 3.3.10.2.

3.3.10.1 Main scales. Instruments with 25 increment vernier scales shall have main scales that are graduated in centimeters, millimeters and half-millimeters. Centimeters shall be numbered consecutively from zero with easily read numerals. Graduation lines representing cm, 1/2 cm, mm, and 1/2 mm shall have lengths in descending order. All graduations shall be of such length as to be easily read. Instruments with 50 increment vernier scales shall have main scales graduated in cm and mm with 1/2 mm graduations at the supplier's option. Each cm graduation within the first 30 cm length of main-scale member shall be accurately spaced with respect to the zero index within 0.008 mm and within 0.008 mm for each additional 30 mm

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increment, total tolerance shall not be accumulated within any 30 cm increment. All mm and 1/2 mm divisions within any 1 cm increment shall be accurately spaced within 0.005 mm.

3.3.10.2 Vernier scales. Vernier scales for metric instruments divided into 25 increments shall have a graduated length of 12 mm or 24-1/2 mm ± 0.005 mm and scales divided into 50 increments shall have a graduated length of 24-1/2 mm or 49 mm, ± 0.005 mm. The spacing and the location of graduations from zero shall be accurate to within 0.005 mm. Graduation lines indicating 0 and 0.1 mm readings shall have extra length and shall be numbered. All graduations and figures shall have sufficient length or height so as to be readily readable.

3.4 Marking. Each instrument shall be marked in a permanent and legible manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined, and the country of origin.

3.5 Heat-treatment. All parts of the instruments requiring heat-treatment shall be heat-treated in such a manner as to obtain the best overall characteristics of the material used in the construction and shall be stable after the final heat-treatment.

3.6 Finish and coating. Except on type II instruments, graduated surfaces of the main-scale member and vernier plate shall have a dull, satin (nonglare) chrome-plated finish. Clamping screws and nuts may be chromium plated or have a colored chemical finish. Other surfaces (other than working or painted), shall have a surface roughness of 30 microinches A.A. or less and may or may not be chromium plated. Type II instruments shall have a polished finish on the main-scale members and vernier plates.

3.7 Instructions. Each caliper or gage shall be accompanied by a clear and concise operational instruction card or leaflet. When specified, card or leaflet shall be enclosed in a waterproof envelope (see 6.2).

3.8 Type I, calipers, vernier. Vernier calipers may be of the open or closed-face type and shall consist essentially of a frame, slide with jaw, vernier scale, fine adjustment clamp, and adjustment screw with nut. The side next to the observer, when the beam is held with the fixed jaw to the left and extending down, shall be known as the front of the instrument and the opposite side, the back of the back of the instrument. However, comparable direct reading openside slide instruments shall be acceptable under this specification. Instruments of this type shall be similar to figure 1 and shall conform to the applicable requirements specified herein.

3.8.1 Frame. The frame shall be composed of a beam and fixed-jaw sections. The main scales shall be located on the front and back sides of the beam section. The measuring contact plane of the fixed jaw shall be square with the slide locating edge of the beam section in the transversal direction within 0.0002 inch in the full width of the contact surface and within 0.0003 inch in the lateral direction in the full length of the contact surface. Any deviation from squareness in the lateral direction shall be in the toe-in direction.

3.8.2 Vernier scale. Each caliper shall be equipped with a vernier scale (or scales, class 3). The graduations shall be in accordance with the requirements specified herein for vernier scales, English or metric, as applicable.

3.8.3 Fine adjustment clamp. A fine adjustment clamp of the slotted type shall be provided.

3.8.4. Fine adjustment screw. A find adjustment screw and nut shall link the fine adjustment clamp and the slide.

3.8.5 Slide with jaw. The movable jaw shall be an integral part of the slide or rigidly attached thereto. Parallelism between the measuring contact planes of the movable and fixed jaws shall be within 0.0003 inch when the slide is locked in position. Any deviation from parallelism within the specified tolerance shall be in the toe-in position.

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3.8.6 Jaws. The jaws shall have nibs so designed as to permit inside as well as outside measurement. The length of the measuring contact planes for external measuring shall be approximately one-half the throat depth. The gaging edge of either jaw shall not be beveled or chamfered. When specified (see 6.2), calipers shall have carbide-tipped jaws.

3.8.7 Tool-setting points. When specified (see 6.2), conical indentures on the frame and slide shall be provided for setting dividers, except the 36, 48 inch and 900 mm calipers. The indentures shall be so located on the calipers as to permit direct reading of the scales. The accuracy in the location of the points, with respect to the zero indexes, shall be within 0.002 inch for English graduated instruments and within 0.05 mm on metric graduated instruments.

3.8.8 Type I, class 1, calipers vernier, English measure (see figure 1). The front of the type I, class 1 calipers shall be graduated for external measuring and the back for internal measuring. The main scales on both the front and back shall be so located as to be direct reading. The frame shall be graduated to the requirements specified herein for main scale "English Graduated Instruments". The vernier scale shall be graduated to the requirements specified herein for "English Graduated Instruments". Instruments of this type and class shall be of ranges as specified in table I. Calipers shall be furnished with the external and internal measuring scales clearly designated.

TABLE I. Type I, class 1 vernier, English measure calipers.

English graduated			
Range from 0	Throat depth (min.)	Nib width jaw closed +0.000 -0.005 inch	Nib length ± 1/16 inch
Inches	Inches	Inch	Inch
6....	1.250	0.250	1/4
9....	2.250	.300	1/4
12....	2.250	.300	1/4
24....	2.250	.300	1/4
36....	2.750	.500	3/8
48....	3.000	.500	3/8

3.8.9 Type I, class 2, calipers; vernier, metric measure (see figure 1). The front of the type I, class 2 calipers shall be graduated for external measuring and the back for internal measuring. The main scales on both the front and back shall be so located as to be direct reading. Instruments of this type and class shall be of ranges as specified in table II. The front of the slide shall be marked "Outside" and the back, "Inside". The vernier scale and the frame shall be graduated to the requirements specified herein for "Metric Graduated Instruments".

TABLE I. Type I, class 1 vernier, metric measure calipers.

Metric graduated			
Range from 0	Throat depth (min.)	Nib width jaw closed +0.000 -0.127 mm	Nib length ± 1 mm
mm	mm	mm	mm
150...	31	6.00	6.4
200...	57	8.00	6.4
300...	57	8.00	6.4
600...	57	8.00	6.4
900...	69	12.00	9.0

3.8.10 Type I, class 3, calipers; vernier, English and metric measure (see figure 1). Type I, class 3 calipers shall be English graduated on the front and metric graduated on the back. The main scales shall be so located as to be direct reading for external measurement. The total nib width (to four decimal places) as measured over both nibs with the jaw closed shall be marked on the slide, both front and back, to provide for measuring internal dimensions. Instruments of this type and class shall be of ranges and general dimensions as shown in table III.

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TABLE III. Type I, class 3 vernier, English and metric measure calipers.

English and metric calibrated					
Range from 0		Throat depth (minimum)		Nib with jaws closed +0.0003	+0.0076
Inches	mm	Inches	mm	Inch	mm
6....	150	1.250	31	0.250	6.35
9....	200	2.250	57	.300	7.62
12....	300	2.250	57	.300	7.62
24....	600	2.250	57	.300	7.62
36....	900	2.750	69	.500	12.70

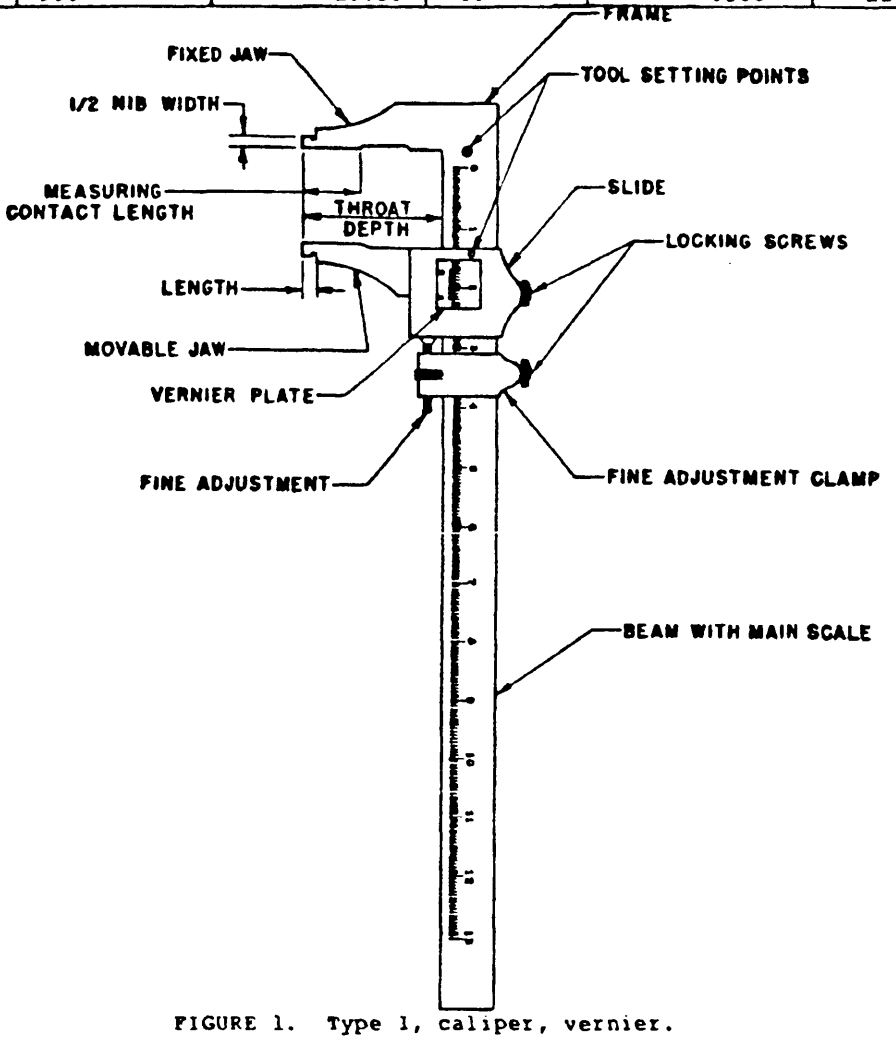


FIGURE 1. Type I, caliper, vernier.

3.9 Type II, calipers; gear tooth vernier. Type II instruments shall be graduated to read from the front. The front of the instrument shall be the side next to the observer when the beam extends to the right and the jaws extend down.

3.9.1 Type II, class 1, style A calipers; gear tooth vernier, English measure 20 to 2 diametral pitch (see figure 2). Style A calipers shall be for use on gears having from 20 to 2 diametral pitch and shall be so marked on the beam section of the frame. Instruments of this type and class shall be composed essentially of a frame with fixed jaw, beam, and column sections; slide with movable jaw, vernier plate, fine adjustment screw with nut, fine adjustment clamp; slide with tongue, vernier plate, fine adjustment screw with nut, and fine adjustment clamp. Calipers of this type, class, and style shall be similar to figure 2.

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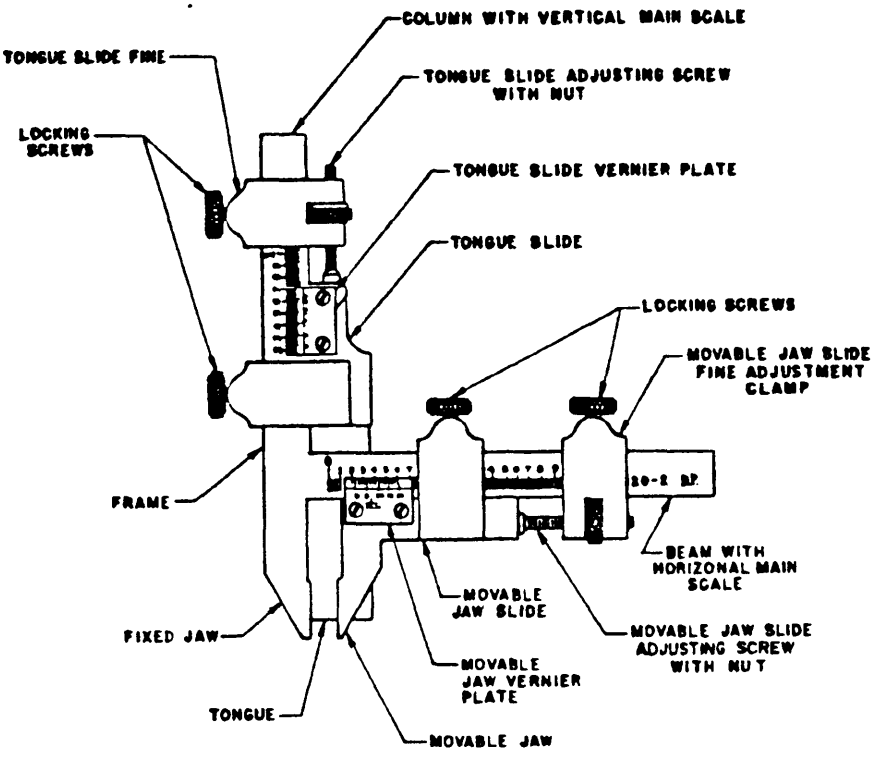


FIGURE 2. Type II, class 1, style A, calipers; gear tooth vernier, 20 to 2 diametral pitch.

3.9.2 Frame. The frame shall consist essentially of three sections; the fixed jaw, beam, and column. The slide locating edges of both the column and beam sections shall be straight from end to end within 0.0002 inch and square with each other within 0.0003 inch for each inch of beam length, total tolerance shall not be accumulated within any 1-inch increment. The beam section shall be slotted in such a manner as to allow the tongue to telescope through the beam section of the jaws.

3.9.3 Slide with jaw. The movable jaw shall be slotted longitudinally for a length, width, and depth necessary to accommodate the tongue, and in no case shall the jaw be slotted through its entire length.

3.9.4 Jaws. The gaging edges of the jaws shall not be beveled or chamfered. When specified (see 6.2), calipers shall have carbide-tipped jaws.

3.9.5 Tongue. The tongue shall be rigidly attached to the tongue slide and shall pass through the slot provided in the beam section of the frame and movable jaw. The contact end of the tongue, when the slide is locked to the column, shall be square with the jaws within 0.0003 inch in the full width of the tongue.

3.9.6 Main scale. There shall be two main scales, one located on the beam and the other on the column sections of the frame. The scale on the beam shall have a graduated length of 2 inches and on the column 1-1/2 inches. Both scales shall be graduated, marked, and shall have the same accuracy requirements as specified herein, except the 0.100-inch increments shall be subdivided into 0.020-inch increments instead of 0.025-inch.

3.9.7 Vernier scales. The vernier scale shall have a graduated length of 0.380-inch and be divided into 0.019-inch increments. The requirements for vernier plate, "English Graduated Instruments" shall apply for accuracy of graduations location, straightness of graduations, and width of graduations.

3.9.8 Type II, class 1, style B calipers; gear tooth, vernier, English measure 10 to 1 diametral pitch (see figure 3). Style B calipers shall be for use on gears

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having from 10 to 1 diametral pitch and shall be so marked on the beam section of the frame. Instruments shall consist essentially of a frame with fixed jaw, beam, and column sections; slide with movable jaw, vernier plate, fine adjustment screw with nut, fine adjustment clamp; tongue with fine adjustment screw with nut, fine adjustment clamp with locknut, and vernier plate. Instruments of this type, class, and style shall be similar to figure 3.

3.9.9 Frame. The frame shall consist essentially of three sections, the fixed jaw, beam, and column. The slide locating edge of the beam and the slotted ways for tongue shall be straight from end to end within 0.0002 inch and be square with each other within 0.0003 inch from each inch of beam length. Total tolerance shall not be accumulated within any 1-inch increment. The column and fixed-jaw section of the frame shall have a groove to guide the tongue and the beam section shall be slotted so as to permit the tongue to telescope through the beam of the jaws.

3.9.10 Slide with jaw. The movable jaw shall be slotted longitudinally for a length, width, and depth to accommodate the tongue. In no case shall the jaw be slotted through its entire length.

3.9.11 Tongue. The tongue shall slide in the slotted guides of the column section and fixed jaw, and shall pass through the slot provided in the beam section and movable jaw. The contact end of the tongue, when locked in place, shall be square with the jaws within 0.0003 inch in the full width of the tongue. The adjusting screw shall be rigidly attached to the top end of the tongue.

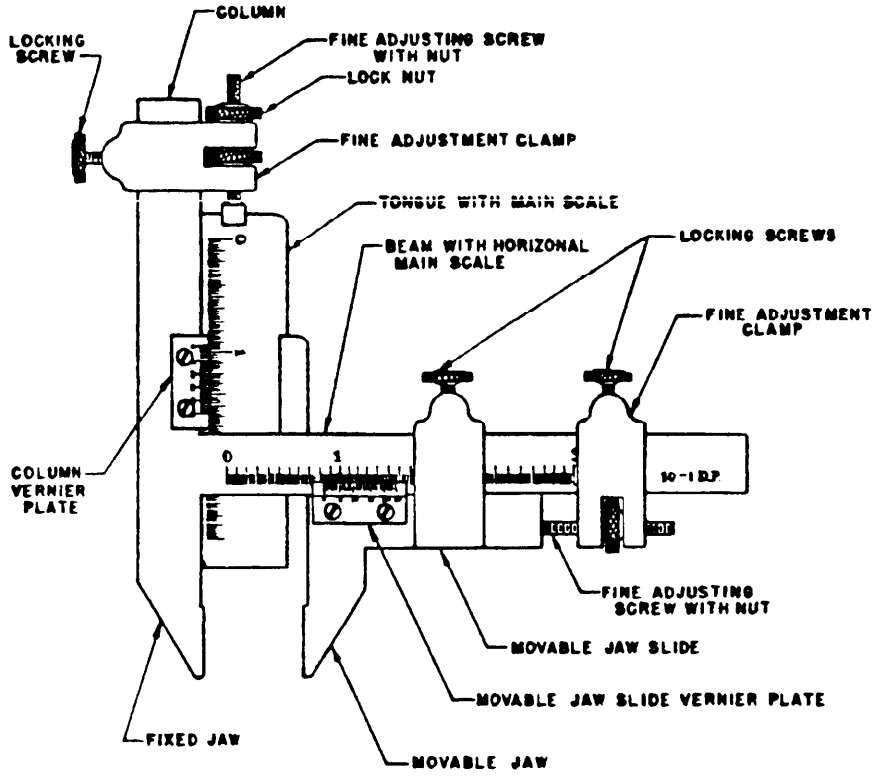


FIGURE 3. Type II, class 1, Style B, calipers; gear tooth vernier, 10 to 1 diametral pitch.

3.9.12 Main scales. There shall be two main scales, one located on the beam and the other on the column of the frame, or the vertical scale may be located on the tongue. The scale on the beam shall have a graduated length of 3 inches. The one on the column or tongue shall have a graduated length of 1-1/2 inches. Both scales shall be graduated, marked, and shall have the same accuracy requirements as specified herein for "English Graduated Instruments".

3.9.13 Vernier scales. One vernier plate shall be mounted on the column of the frame or on the tongue, depending upon the location of the vertical main scale; the

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other shall be mounted conventionally. Both vernier scales shall be graduated as specified herein, for "English Graduated Instruments."

3.10 Type III, gages; depth, vernier (see figure 4). Type III depth gages shall be composed essentially of a head with base, slide, and vernier plate, and blade. The front of the instrument shall be the side next to the observer when the instrument is resting on its base with the vernier plate to the right side. Instruments of this type shall be similar to figure 4. The head and base shall be integral. The slide, vernier plate adjusting clamp, and screws shall conform to the applicable requirements for type 1, class 1 (English graduated) calipers. When specified (see 6.2), a fine adjustment clamp and screw shall be provided on type III gages.

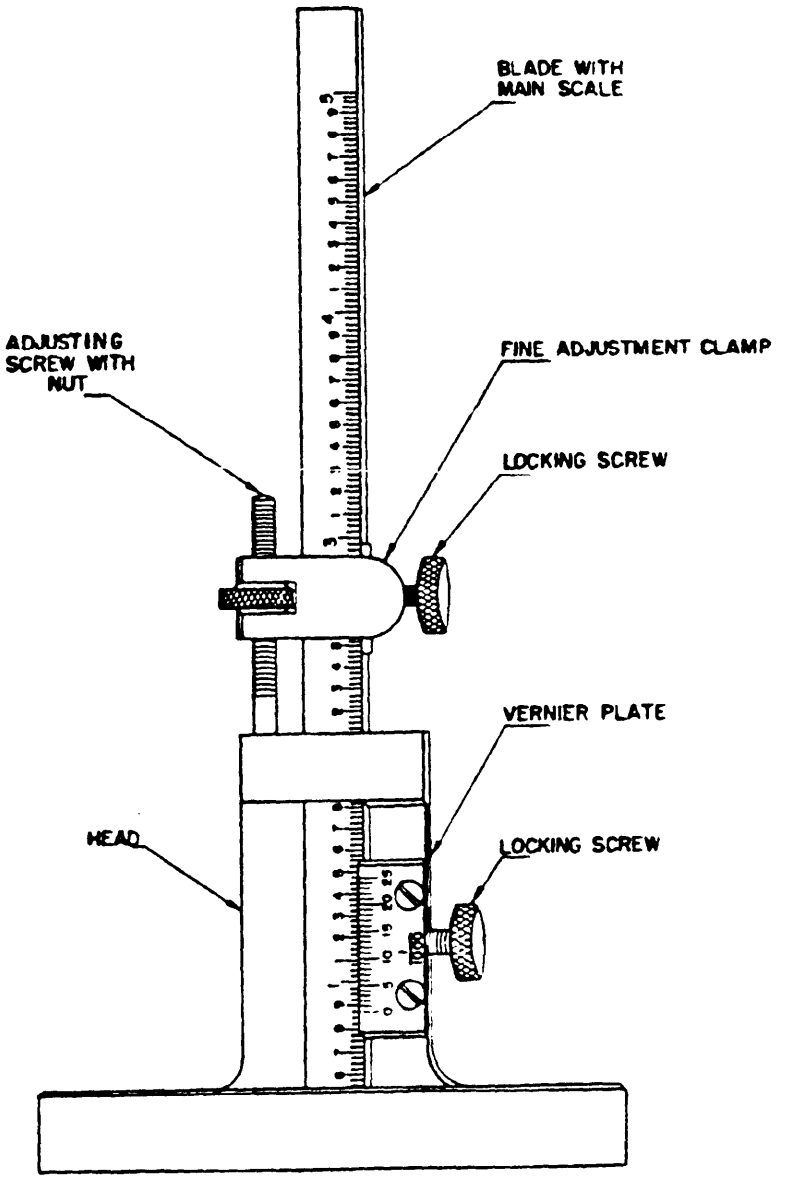


FIGURE 4. Type III, gage; depth, vernier (with fine adjustment clamp and screw)

3.10.1 Base. Dimensions of the base (extended portion of the head), shall be not less than 3/16 nor more than 5/16 inch in thickness (front to back), and not less than 2-5/8 inches in length. The measuring face shall have a hardness of 60

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to 65, inclusive, on the Rockwell C scale, and shall be planed within 0.0001 inch and have a finish such that the surface roughness does not exceed 16 microinches A.A.

3.10.2 Blades. Blades shall be of tool steel, hardened and ground, designed to fit the head, and provide maximum accuracy. Blades shall be of the length hereinafter specified for class and size. Blades shall be graduated as specified for each class.

3.10.3 Finish. The base and fine adjusting clamp shall have a dull chrome (nonglare) finish, a polished finish, or shall have a chemically colored finish at the manufacturer's option.

3.10.4 Type III, class 1, gages; depth, vernier, English measure (see figure 4). Class 1 instruments shall be English graduated on the front side only and shall conform to the requirements specified herein for "English Graduated Instruments". They shall have one blade at least 6 inches long with a 3-1/2 inch range (minimum) and another blade at least 12 inches long with a 9-1/2 inch range.

3.10.5 Type III, class 2, gages; depth, vernier, metric measure (see figure 4). Class 2 instruments shall be metric graduated on the front side only and shall conform to the requirements specified herein for "Metric Graduated Instruments". They shall have one blade at least 15-1/4 cm in length with a range of 88 mm, and another blade at least 30-1/2 cm in length with a range of 238 mm.

3.10.6 Type III, class 3, gages; depth, vernier, graduated English and metric measure (see figure 4). Class 3 instruments shall have blades with English and metric graduations conforming to the requirements for "English" and "Metric Graduated Instruments". One blade shall be at least 6 inches in length with a range of 3-1/2 inches on the English graduated side and 88 mm on the metric side, and shall have another blade at least 12 inches long with a range of 9-1/2 inches on the English graduated side and 238 mm on the metric side.

3.11 Type IV, gages; vernier, height. Type IV gages shall consist essentially of a frame, base, slide with arm, scribing point, vernier scale, fine adjustment mechanism, and accessory clamp. Type IV gages shall conform to table IV and be similar to figure 5, but not necessarily identical in design to figure 5.

TABLE IV. Type IV, classes 1, 2, and 3 vernier height gages.

Size		Minimum 1/ capacity range		Base	
				Length min.	Width min.
Inches	Cm	Inch	Cm	Inches	Inches
6	15	0-6	0-15	2-3/4	1
12	30	1-1/4-12	3-30	3-1/2	1-1/2
18	45	1-1/2-18	4-45	4-7/16	2
24	60	2-24	5-60	5-3/4	2-3/4
36	90	2-36	5-90	5-3/4	3
48	120	0-48	0-120	7-1/2	5-1/2
60		0-60		10	6-1/2
72		0-72		10	6-1/2

1/ Even though minimum capacities greater than zero are shown in table IV, each type IV gage shall have an offset arm or scribing point which shall line up with the bottom edge of the base. The offset arm or scribing point may be integral or an attachment.

3.11.1 Frame. The frame shall consist of column and base, formed or assembled to form a permanently integral unit. There shall be no looseness or play at the point of joining.

3.11.1.1 Column. The column shall be of such shape and size as to provide rigid support to the slide within the accuracy requirements specified herein. The column shall be graduated for each class as hereinafter specified. The column shall be of tool steel.

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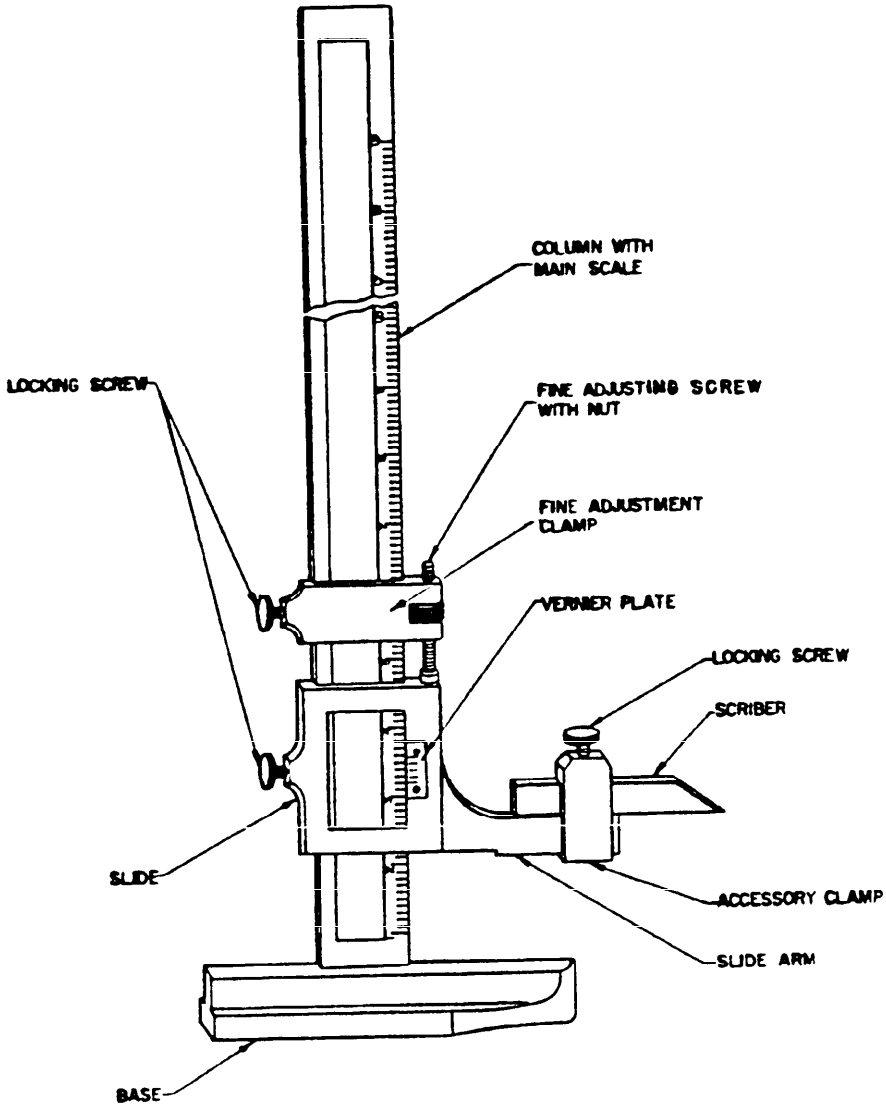


FIGURE 5. Type IV, gages; vernier, height.

3.11.1.2 Base. The base member shall be hardened to not less than 60 on the Rockwell C scale. The bottom surface of the base shall be relieved in the center so that surface contact is made at the outer edges. The relieved portion shall not extend nearer the outer edge than 5/32 inch. The relieved portion of the base shall be vented.

The bottom surface of the base shall be ground and lapped square with the slide locating edge on the column within 0.0004 inch for each foot of column height. Total tolerance shall not be accumulated in any one foot, but shall be evenly distributed over the entire column height. The contact surface on the base shall have not less than 75 percent bearing.

3.11.2 Slide. The slide shall conform to the general slide requirements specified herein.

3.11.2.1 Vernier scale. The vernier scale shall conform to the general requirements for vernier plate specified herein, except in lieu of the plate, a glass vernier window may be mounted directly over the main scale. The reverse side of the glass shall be engraved or etched. The length of graduations, width of graduations, spacing, straightness of graduations, and graduated length of the glass

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vernier window shall be in accordance with the requirements specified herein for vernier scales. The glass vernier window shall be mounted through the use of four or more screws and shall be provided with not less than 0.010 inch adjustment for purposes of instrument calibration.

3.11.2.2 Fine adjustment mechanism. The fine adjustment mechanism shall be as herein specified or be partially mounted in and adjustable from the base or be contained in the slide.

3.11.2.3 Slide arm. The slide arm shall be attached rigidly to the slide and shall be provided with an offset scribing point (either integral or an accessory) which shall be in alignment with the bottom of the base when the height gage is set at its minimum graduation. The gaging surfaces of the arm shall be lapped parallel to bottom of the base within 0.0003 inch when the slide is in locked position.

3.11.2.4 Accessory clamp. An accessory clamp of the slotted type shall be provided for mounting scriber points on the slide arm.

3.11.3 Type IV, class 1, gages; height. Type IV, class 1 gages shall have main scales and verniers with English graduations in accordance with the requirements specified herein for "English Graduated Instruments".

3.11.4 Type IV, class 2, gages; height. Type IV, class 2 gages shall have main scales and verniers graduated in the metric system in accordance with the requirements specified herein for "Metric Graduated Instruments".

3.11.5 Type IV, class 3, gages; height. Type IV, class 3 gages shall have main scales and verniers graduated in both the English and metric systems in accordance with the requirements specified herein for each, respectively.

3.11.6 Depth gage attachment. When specified (see 6.2), each height gage shall be furnished with a depth gage attachment and a clamp for attachment to the slide arm.

3.12 Cases. Unless otherwise specified (see 6.2) each vernier instrument shall be furnished in a wooden box or leatherette case. The box or case shall have a hinged cover and clasp or a spring-loaded hinge closure. The instrument shall rest in recesses or be secured by supporting brackets to prevent damage to any part thereof during ordinary usage and handling. All internal surfaces or supports coming in contact with the instrument shall be lined with cushioning material to prevent damage to instrument. Wooden boxes shall be smoothly finished; contain no chipped or splintered wood; have tight glue, nail, or screwjoints; and be provided with a protective coating of lacquer or varnish. Leatherette cases shall be smooth with no rough finished, peeling or sharp edges.

3.13 Predelivery calibration. Each instrument shall be calibrated by adjustment of the vernier, to read the size of the gaging used for setting purposes, prior to delivery.

3.14 Workmanship. The instruments and cases shall be free from all defects or blemishes that might affect their serviceability, durability, or appearance. All graduation lines and markings specified shall be distinct and easily readable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1 above, the contractor is responsible for insuring that materials and components used were

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manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 Sampling procedures. Sampling procedures shall be in accordance with MIL-STD-105. Data for sampling shall be as stated in table V.

TABLE V. Sampling data.

Category	Sample unit	Inspection level	Acceptable quality level	AQL expressed in terms of	Reference
Visual examination	1 ea.	II	1.5 Major 2.5 Minor	Percent defective	4.3.1
Dimensional examination	1 ea.	II	1.5 Major 2.5 Minor	Percent defective	4.3.2
Testing	1 ea.	II	1.5	Percent defective	4.4
Preparation for delivery	one container	S-2	4.0	Defects per hundred units	4.6

4.3 Examination.

4.3.1 Visual examination. Each sample unit shall be examined for any nonconformance in design, material, finish, coating, construction, workmanship, and marking. Defects are listed in table VI.

TABLE VI. Classification of defects.

Categories	Defects
Major:	
101	Type, class, style, or range not as specified.
102	Caliper (or gage) incomplete, parts missing, defective or improperly assembled, evidence of rust, corrosion, cracks, splits, sharp edges, and burrs.
103	Slide or tongue (with locking screws loosened) sticks or binds, or evidence of excessive looseness (lateral shake) (types I, II, and IV only).
104	Blade (with locking screws loosened) sticks or binds, or evidence of excessive looseness (lateral shake) (type III only).
105	Locking screws defective, heads not knurled, or screws not standard fine thread series.
106	Fine adjustment screw defective, not securely fastened to the slide (or tongue), adjusting nuts or knobs not knurled, nuts fail to operate along the screws without sticking or binding, or nuts and screws not standard fine thread series.
107	Tongue not rigidly attached to the slide or beam not slotted to accommodate the tongue (type II only).
108	Vernier plates defective, not securely fastened, plate overlap on main scales less than the specified minimum, graduated edge not beveled, or amount of longitudinal adjustment not as required.
109	Tool setting points (if applicable), missing or not properly located (type I only).
110	Moveable jaw not slotted for the length, width, and depth required to accommodate the tongue, or movable jaw is slotted through its entire length (type II only).
111	Blades not interchangeable or length of blades not as specified (type III only).

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TABLE VI. Classification of defects. (continued)

Categories	Defects
Major:	
112	Graduations (main and vernier scales) missing, incomplete, illegible, not machine cut (or etched), edges not well defined, or graduations not as specified.
113	Throat depth less than the specified minimum (type I only).
114	Nib width exceeds the specified maximum (type I only).
115	Nib length not as required (type I only).
116	Exact nib width marking (front and back scales) missing, illegible, or incorrect (type I, class 3 only).
117	Caliper unstable (flexible) when used to measure at its maximum range (types I and II only).
118	Gage unstable (flexible) when used to measure at its maximum range (types III and IV only).
119	Bottom surface of base not relieved in the center, relieved portion not vented, or bearing surface contact width less than the specified minimum (type IV only).
120	Arm not rigidly attached to the slide, accessory clamp defective, or not as specified (type IV only).
Minor:	
201	Thickness of head not within the specified maximum and minimum values (type III only).
202	Length of head less than the specified minimum (type III only).
203	Length and width of base (as applicable) not within the specified minimum values (type IV only).
204	Marking, manufacturer's name or trademark or country of origin missing, illegible, incorrect, or not permanent.
205	Operating instruction leaflet missing or not enclosed in a waterproof envelope.
206	Case defective, or fails to hold parts in such manner as to preclude damage during handling.

4.3.2 Dimensional examination. Each sample unit shall be examined for any nonconformance with dimensional requirements.

4.4 Testing. Each sample unit shall be tested in accordance with 4.5.1 through 4.5.8.

4.5. Test procedures.

4.5.1 Hardness. Where practicable, hardness shall be determined from a minimum of three readings, taken at points representative of the hardened area or areas concerned. Any individual reading outside the applicable range of hardness specified in section 3 shall be cause for rejection. Hardness tests shall be conducted using low loads to prevent destruction of the part under test and the results converted to the Rockwell "C" scale.

4.5.2 Surface roughness. Surface roughness shall be determined at representative spots by commercially recognized instruments for surface roughness measurement.

4.5.3 Planeness. Planeness shall be checked with an optical flat where the surface so permits, or by means of a precision indicator and precision reference surface, the test piece being supported as required by toolmaker's angles or other means of adequate accuracy.

4.5.4 Angular trueness of reference surfaces. Angular trueness of reference surfaces shall be checked by means of a precision square and feeler gage, precision angle blocks, precision dividing head, autocollimator, or method of equivalent accuracy.

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4.5.5 Graduations. Graduations shall be checked for straightness of edges, width, spacing, parallelism, and perpendicularity to edge of graduated surface by means of a toolmaker's microscope or a method of comparable accuracy. Depth of graduations shall be checked by measuring the height of the ridges formed in a wax impression or by a method of comparable accuracy.

4.5.6 Edge straightness. Edge straightness shall be tested with straight edges of known accuracy, with a precision indicator and reference surfaces such as a precision surface plate, with an autocollimator or method of comparable accuracy.

4.5.7 Profiles. Profiles shall be tested by means of precision templates and feeler gages, where applicable, or by means of comparable accuracy.

4.5.8 Error in indicated measurement. Error in indicated measurement shall be determined by securing the gage in correct relation to a precision surface and checking a series of readings against precision end measures such as gage blocks. The following denominations of length standards shall be included for inch and millimeter reading gages: 0.105, 0.500, and 1.000 inch or 2.5, 15.0,. and 25 millimeters as applicable.

4.6 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing, and marking comply with the requirements in section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with PPP-P-40.

6. NOTES

6.1 Intended use. Type I calipers are intended to be used for accurate measuring of both inside and outside diameters. Type II calipers are intended to be used in measuring gear tooth thicknesses and are applicable for measuring hobs, forms, and thread tools. Type III gages are designed for measuring depth of holes, slots, and recesses. Type IV gages are used for precision height measurements.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class, style, and size of instrument (see 1.2.1 and applicable tables).
(c) Specify if tool setting points are required (see 3.8.7).
(d) Whether waterproof envelope is required (see 3.7).
(e) Whether jaws shall be carbide tipped (see 3.8.6 and 3.9.4).
(f) Whether depth gage attachment is required (see 3.11.6).
(g) Whether a fine adjustment screw and clamp shall be supplied with type III gages (see 3.10).
(h) Case requirements, if different (see 3.12).

MILITARY INTERESTS:

NONE: DoD has determined that no military activity has an official interest in this Federal specification.

PREPARING ACTIVITY

GSA/FSS

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